Marketing Analytics

Chapter 6 · Mechanisms & Heterogeneity

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Before we start

Paper quiz on Chapter 5, Part II *(§5.7–5.10, 8 min)*

- Individual, closed book
- Counts toward participation

Assignment 3 is running

- Deadline: Group A Sun, Nov 22 · Group B Wed, Nov 25
- Debrief at the start of next session

Today's session

6.1 The average effect hides the interesting part
6.2 Interactions and **marginal effects**
6.3 Continuous × categorical
6.4 Continuous × continuous *(short)*
6.5 Categorical × categorical: a **2 × 3 experiment**
6.6 Pitfalls: subgroup fishing & power
6.7 **Mediation**: why does it work?
6.8 Good and bad controls *(short)*
6.9 Your group project

> **Driving question:** *Why — and for whom — does it work?*
>
> One new ingredient: the **product of two variables**.

6.1 The average effect hides the interesting part

Smart Mix, second wave

- **1,200 users**, randomized to Smart Mix or control
- This time with **covariates**: plan (free/Premium), prior listening, tenure, device
- And one more variable: **new artists discovered** during the four weeks
- Average effect: **+3.7 hours** (95% CI 3.1–4.3)

The product team asks two more questions:

1. **For whom** does it work — free or Premium users? light or heavy listeners?
2. **Why** does it work — does it make users discover new music?

Moderation (for whom): the effect *depends on* a third variable. **Mediation** (why): the effect *runs through* a third variable. Easy to confuse — a moderator changes *how large*, a mediator is *part of the path*.

6.2 Interactions and marginal effects

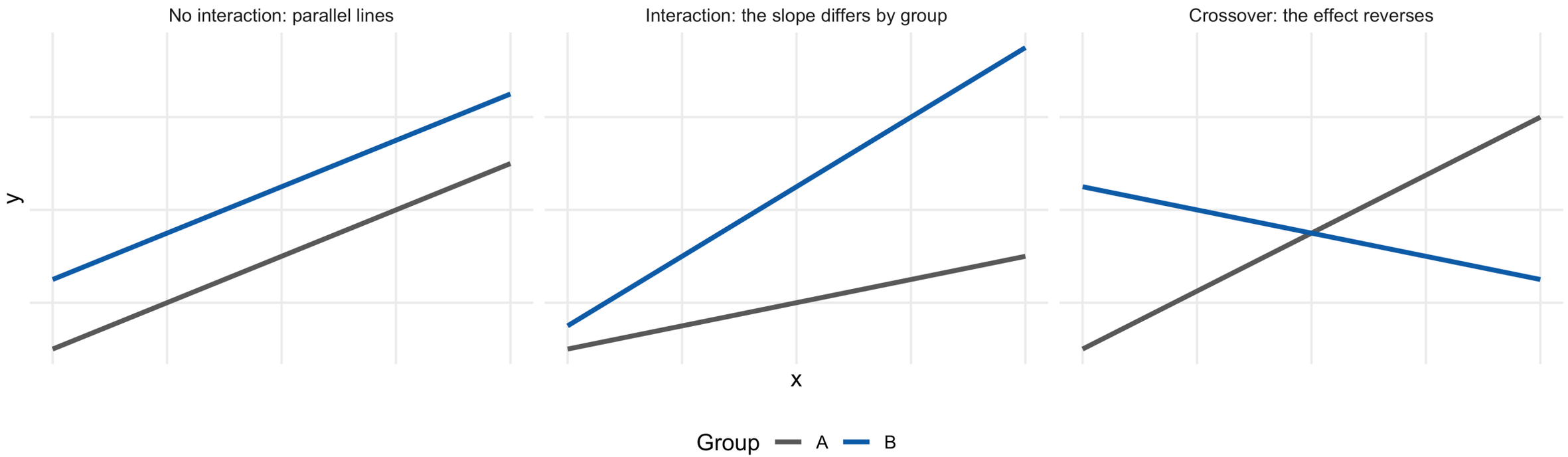
What an interaction term does

$$y = \beta_0 + \beta_1 x + \beta_2 z + \beta_3 (x \times z) + \varepsilon \qquad \Rightarrow \qquad \frac{\partial y}{\partial x} = \beta_1 + \beta_3 z$$

- The **marginal effect** of x depends on z
- β_1 = effect of x **when** $z = 0$ (a *conditional* effect, not a "main effect")
- β_3 = how much the effect of x changes per unit of z
- Symmetric: effect of z = $\beta_2 + \beta_3 x$

Two rules: (1) if zero is not a meaningful value of z, β_1 is not meaningful either → center (§6.4). (2) **Always keep both main terms**: in R, `x * z` = `x + z + x:z`.

Parallel, diverging, crossing



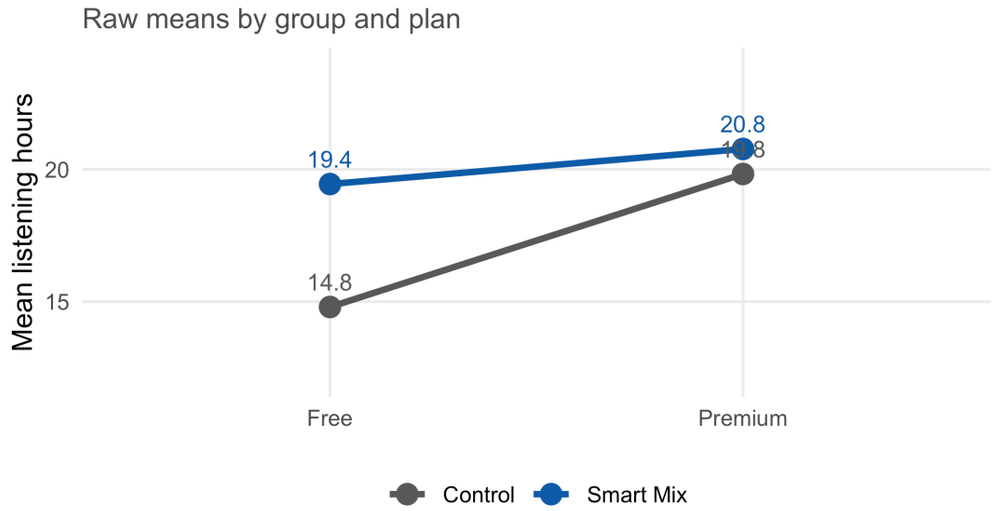
With a group indicator as z: **no interaction** = parallel lines (same slope, different level) · **interaction** = different slopes · **crossover** = the effect of x reverses. β_3 is simply the **difference between the slopes**.

6.3 Continuous × categorical

Face 1: a treatment effect that differs between groups

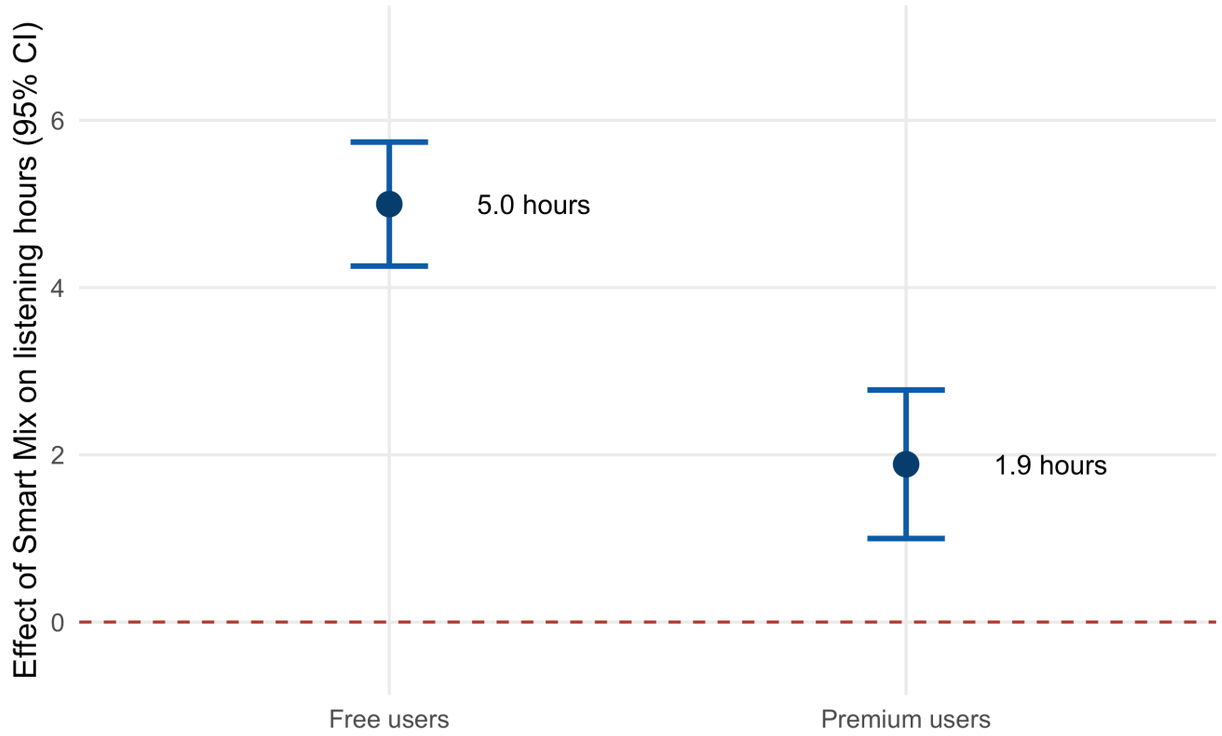
```r
model_plan <- lm(hours ~ smart_mix * premium + prior_hours + tenure, data = wave2)
coef_table(model_plan)
```

	Estimate	Std. Error	t value	p-value
(Intercept)	1.560	0.469	3.33	< 0.001
smart_mix	4.999	0.378	13.21	< 0.001
premium	2.685	0.431	6.22	< 0.001
prior_hours	0.752	0.017	43.97	< 0.001
tenure	0.028	0.009	3.26	0.001
smart_mix:premium	−3.111	0.590	−5.27	< 0.001



`smart_mix` = effect **for free users** (premium = 0): **5.0 h** · `smart_mix:premium` = how much the effect **differs** for Premium: **−3.1 h** (the heterogeneity test) · effect for Premium = 5.0 − 3.1 = **1.9 h** · `premium` = plan difference **in the control group**.

Both effects with their uncertainty



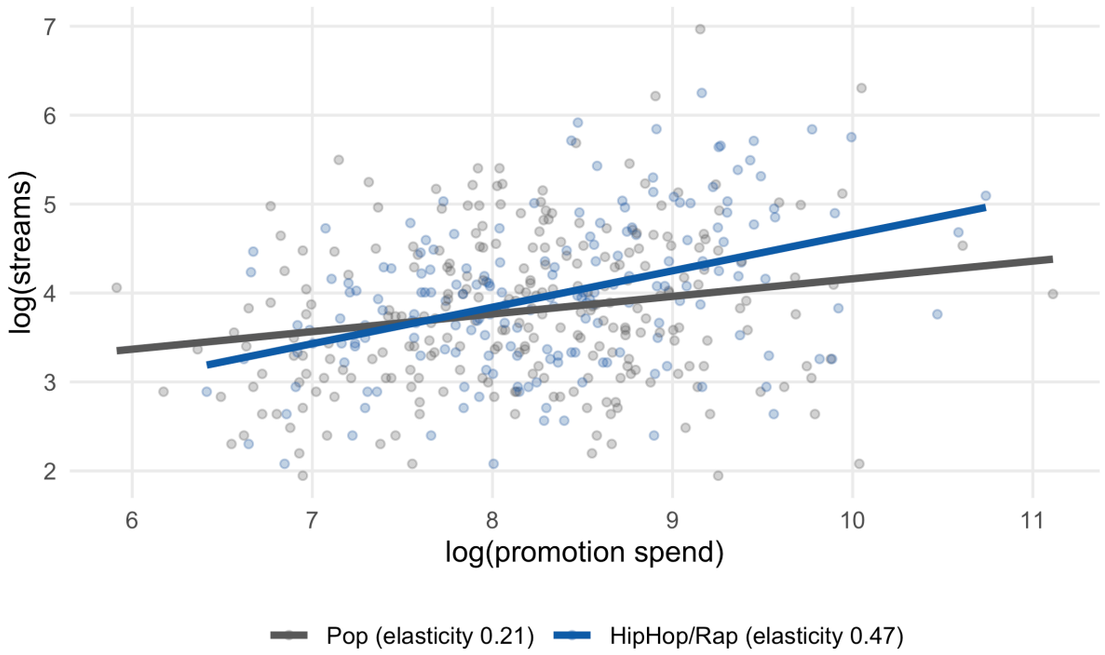
- The CI of the Premium effect is **not** in the output — it combines two coefficients and their covariance (`emmeans` does it for you)
- Actionable: promote Smart Mix to **free users first**

> The interaction coefficient tests whether the effects **differ**; the two marginal effects tell you **what they are**. Report both.

Face 2: a slope that differs between groups

```r
model_genre <- lm(log(streams) ~ log(spend) * genre + log(followers) + plac
                  data = promo)
coef_table(model_genre)
```

	Estimate	Std. Error	t value	p-value
(Intercept)	−3.648	0.321	−11.36	< 0.001
log(spend)	0.206	0.033	6.17	< 0.001
genreHipHop/Rap	−2.058	0.430	−4.79	< 0.001
genreRock	0.356	0.654	0.54	0.587
genreElectro/Dance	0.021	0.616	0.03	0.973
log(followers)	0.507	0.015	33.46	< 0.001
placements	0.122	0.011	11.42	< 0.001
log(spend):genreHipHop/Rap	0.268	0.052	5.18	< 0.001
log(spend):genreRock	−0.053	0.079	−0.67	0.504
log(spend):genreElectro/Dance	0.005	0.076	0.07	0.946



Same model, different question: `log(spend)` = elasticity for **Pop** (reference); `log(spend):genreHipHop/Rap` = **+0.27** → HipHop/Rap elasticity ≈ **0.47**. Rock and Electro: no evidence of a different slope. Genre "main effects" = level differences at **1 euro** of spend.

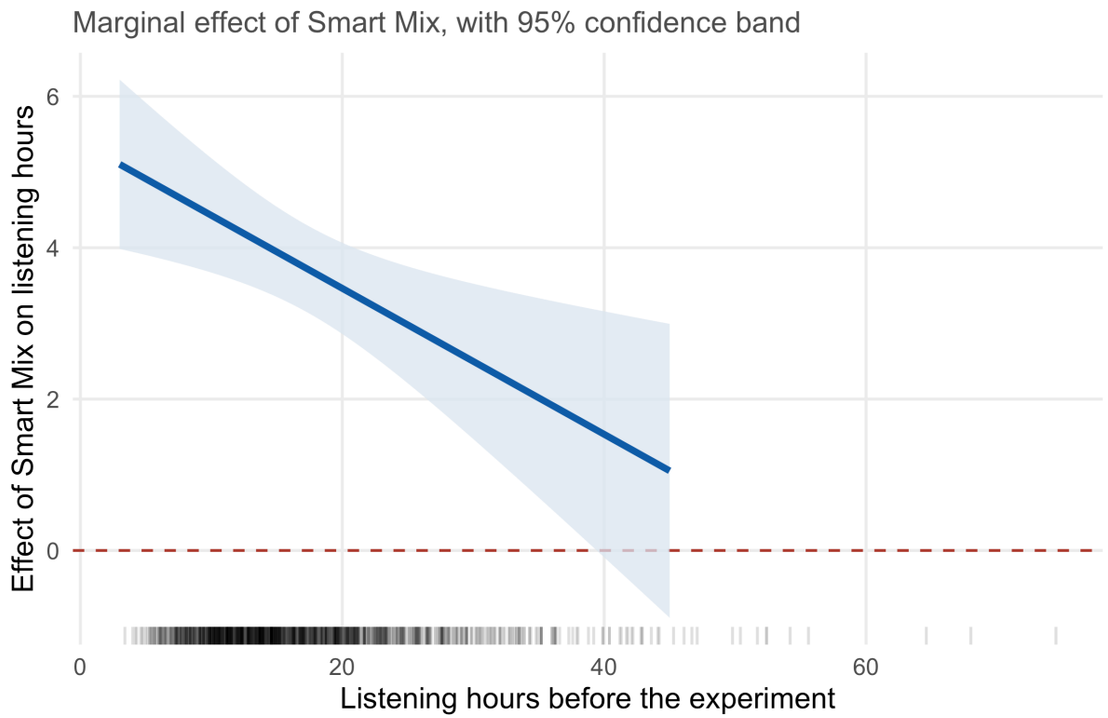
Moderation by a continuous variable

```
model_prior <- lm(hours ~ smart_mix * prior_hours + premium + tenure, data
coef_table(model_prior)
```

	Estimate	Std. Error	t value	p-value
(Intercept)	1.446	0.532	2.72	0.007
smart_mix	5.391	0.660	8.16	< 0.001
prior_hours	0.797	0.023	34.43	< 0.001
premium	1.041	0.301	3.46	< 0.001
tenure	0.027	0.009	3.15	0.002
smart_mix:prior_hours	−0.096	0.034	−2.82	0.005

Marginal effect of Smart Mix = 5.39 − 0.096 × prior hours:

- 10 hours before → **4.4 h** · 30 hours → **2.5 h**
- `smart_mix` alone = effect at **zero** prior hours — a user who hardly exists



With a continuous moderator, report the effect at **meaningful values** (10th / 50th / 90th percentile) or as a **curve with its confidence band** — never the coefficient at zero.

Your turn (5 min)

Fit the interaction of Smart Mix with **device**:

```
model_device <- lm(hours ~ smart_mix * device + prior_hours + premium + tenure, data = wave2)
coef_table(model_device)
```

For which device is the effect of Smart Mix largest? Compute the effect for each device from the coefficients. Are the differences significant?

6.4 Continuous × continuous

Spend × followers: the problem with “main effects”

Uncentered:

```
m_cc <- lm(log(streams) ~ log(spend) * log(followers) +
             placements + genre, data = promo)
coef_table(m_cc)[2:3, ]
```

```
               Estimate Std. Error t value p-value
log(spend)       -0.896      0.183   -4.90 < 0.001
log(followers)   -0.422      0.143   -2.96   0.003
```

A **negative** spend elasticity? It is the elasticity for an artist with **one follower** (log = 0).

Centered at the means:

```
promo_c <- promo |> mutate(ls = log(spend) - mean(log(spend)),
                           lf = log(followers) - mean(log(followers)))
m_ccc <- lm(log(streams) ~ ls * lf + placements + genre, data = promo_c)
coef_table(m_ccc)[c(2, 3, 8), ]
```

```
      Estimate Std. Error t value p-value
ls       0.299      0.022   13.31 < 0.001
lf       0.508      0.015   33.87 < 0.001
ls:lf    0.113      0.017    6.53 < 0.001
```

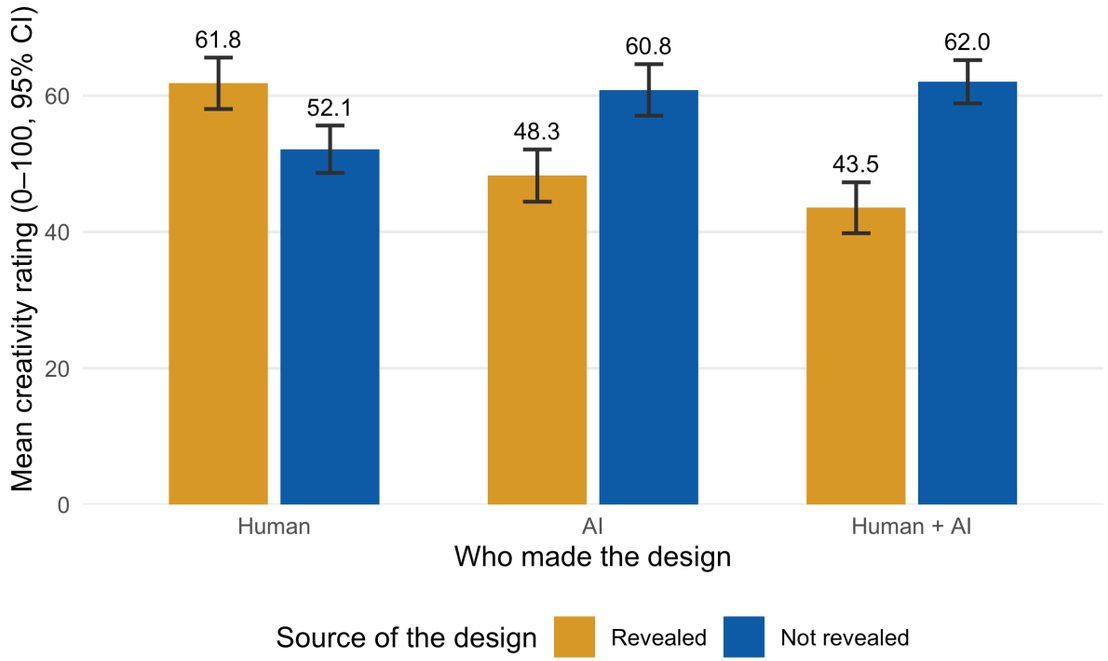
Now `ls` = elasticity for an **average** artist; the interaction is unchanged.

> Centering changes the main effects (now “at the mean of the other variable”), **not** the interaction or the fit. Communicate with a **plot of the marginal effect** — book §6.4 — not with the table.

6.5 Categorical × categorical: a 2 × 3 experiment

Are AI-made ads judged as creative as human-made ads?

- **360 respondents** rate the creativity of an ad (0–100)
- Factor 1 — **source** of the design: Human · AI · Human + AI
- Factor 2 — **disclosure**: source revealed · not revealed
- 2 × 3 = **6 cells**, 60 respondents each, randomized



Revealed: human designs win. Not revealed: AI and hybrid designs are rated **as creative** — slightly higher. **Crossing lines = the effect of the source depends on disclosure.**

The regression: six coefficients, six cell means

```
model_ads <- lm(creativity ~ source * disclosure, data = ads)
coef_table(model_ads)
```

	Estimate	Std. Error	t value	p-value
(Intercept)	61.800	1.854	33.33	< 0.001
sourceAI	−13.533	2.622	−5.16	< 0.001
sourceHuman + AI	−18.250	2.622	−6.96	< 0.001
disclosureNot revealed	−9.667	2.622	−3.69	< 0.001
sourceAI:disclosureNot revealed	22.233	3.708	6.00	< 0.001
sourceHuman + AI:disclosureNot revealed	28.150	3.708	7.59	< 0.001

Coefficient	Meaning (reference: Human, Revealed)
`(Intercept)` 61.8	Human designs, source revealed
`sourceAI` −13.5 · `sourceHuman + AI` −18.3	AI / hybrid designs, *when revealed*, vs. human
`disclosureNot revealed` −9.7	*Human* designs: not revealed vs. revealed
`sourceAI:…` +22.2 · `sourceHuman + AI:…` +28.2	The **extra** gain of AI / hybrid designs when not revealed, beyond the two main effects

Step 1: is there an interaction at all?

```
1 anova(model_ads)
```

```
Analysis of Variance Table

Response: creativity
                  Df Sum Sq Mean Sq F value  Pr(>F)
source             2   1055     527    2.56   0.079 .
disclosure         1   4572    4572   22.17 3.6e-06 ***
source:disclosure  2  13218    6609   32.04 1.6e-13 ***
Residuals        354  73021     206
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

> The ANOVA table tests each factor and the **interaction as a whole** (two coefficients at once). Interaction highly significant → the effect of the source depends on disclosure → **do not interpret the main effects on their own**. (Unbalanced cells: `car::Anova(model_ads, type = 2)`.)

Step 2: which cells differ? Estimated marginal means

```r
library(emmeans)
emm <- emmeans(model_ads, ~ source | disclosure)
pairs(emm)
```

```
disclosure = Revealed:
 contrast             estimate   SE  df t.ratio p.value
 Human - AI              13.53 2.62 354   5.161  <.0001
 Human - (Human + AI)    18.25 2.62 354   6.960  <.0001
 AI - (Human + AI)        4.72 2.62 354   1.799  0.1715

disclosure = Not revealed:
 contrast             estimate   SE  df t.ratio p.value
 Human - AI              -8.70 2.62 354  -3.318  0.0029
 Human - (Human + AI)    -9.90 2.62 354  -3.775  0.0005
 AI - (Human + AI)       -1.20 2.62 354  -0.458  0.8910

P value adjustment: tukey method for comparing a family of 3 estimates
```

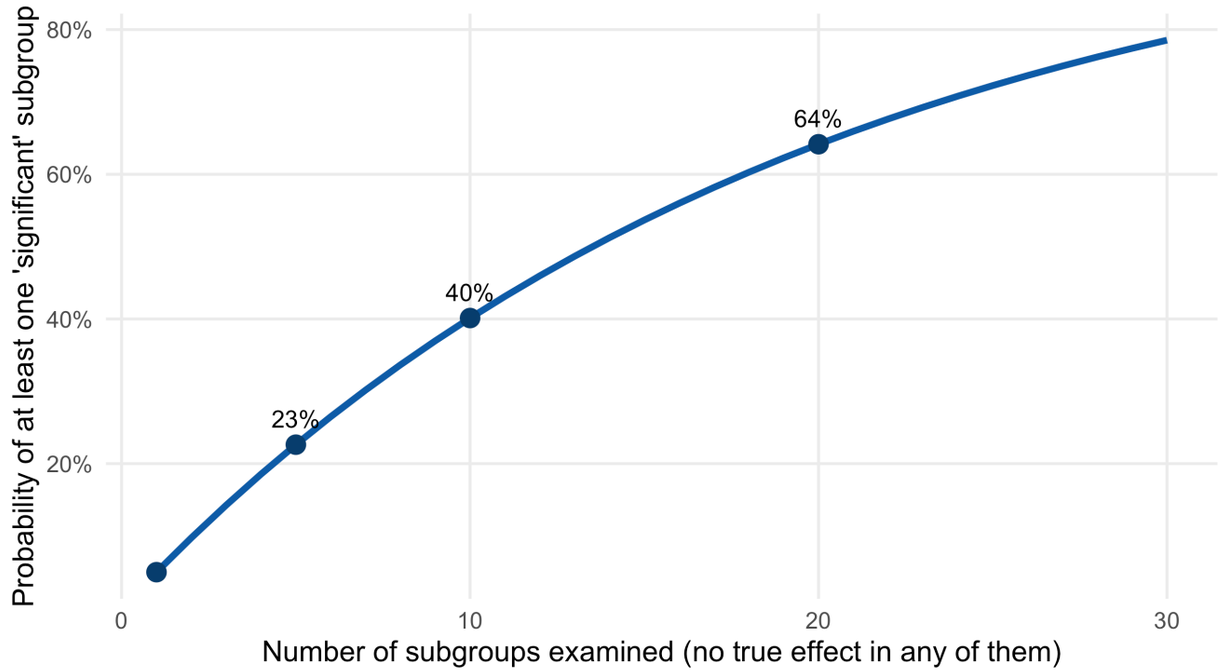
> From “there is an interaction” to “**where exactly** are the differences”: revealed → human beats AI (14 points) and hybrid; not revealed → AI and hybrid beat human by 9–10 points. *p*-values **Tukey-adjusted** within each disclosure condition — the two-step logic of Chapter 4, with one extra dimension.

Your turn (5 min)

1. Rebuild the cell mean for **hybrid** designs with the source **not revealed** from the coefficients of `model_ads`. Check it against the table of cell means.
2. Which **single number** in the regression output answers: “does disclosure change the rating of hybrid designs *more* than it changes the rating of human designs?”

6.6 The pitfalls of looking for heterogeneity

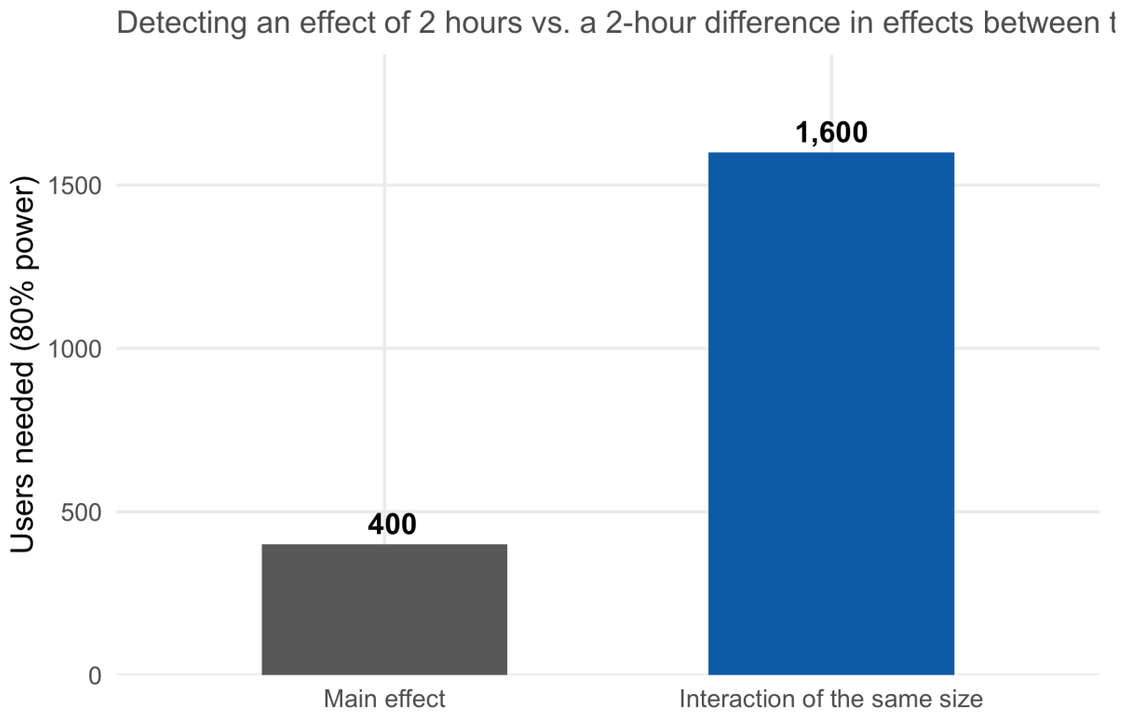
Subgroup fishing



- Plan × device × tenure × age × country … = dozens of subgroups
- Testing each = **multiple testing** (Chapter 4)
- 20 subgroups, no heterogeneity: **64%** chance of a "significant" one
- Live demo: subgroup-fishing widget, book §6.6

> "Smart Mix works particularly well for Android users aged 25–34" after twenty subgroups = **noise with a story**.

Interactions need large samples



- SE of a *difference between two effects* = **2 ×** SE of the overall effect
- Same power → **4 × the sample** for an interaction as large as a main effect
- And interactions are usually **smaller**
- Underpowered: miss real heterogeneity, **exaggerate** what you find

Pre-register the moderators you expect and test those. Everything else is exploratory — a hypothesis for the *next* experiment. Not significant → “no evidence that the effect differs”, not “the effect is the same”.

6.7 Mediation: why does it work?

Total = direct + indirect

Three regressions

```r
model_m <- lm(new_artists ~ smart_mix + prior_hours + premium, data = wave2)          # path a
model_y <- lm(hours ~ smart_mix + new_artists + prior_hours + premium + tenure, data = wave2)   # paths b and c'
a <- coef(model_m)["smart_mix"]; b <- coef(model_y)["new_artists"]; c_direct <- coef(model_y)["smart_mix"]
c(a = unname(a), b = unname(b), direct = unname(c_direct), indirect = unname(a * b), total = unname(c_direct + a * b))
```

```
     a        b   direct indirect    total
3.6185   0.3909   2.3063   1.4146   3.7209
```

- **a** = 3.6 more new artists with Smart Mix
- **b** = 0.39 hours per new artist, holding Smart Mix constant
- **c′** = 2.31 hours direct — down from the total 3.72

> Indirect effect a × b = **1.42 hours = 38%** of the effect runs through discovery. Discovery is **an** important mechanism — not the only one.

Uncertainty of the indirect effect: bootstrap

```r
set.seed(1)
boot_indirect <- replicate(2000, {
  d_boot <- wave2[sample(nrow(wave2), replace = TRUE), ]
  coef(lm(new_artists ~ smart_mix + prior_hours + premium, data = d_boot))["smart_mix"] *
    coef(lm(hours ~ smart_mix + new_artists + prior_hours + premium + tenure, data = d_boot))["new_artists"]
})
quantile(boot_indirect, c(0.025, 0.975))
```

```
  2.5%  97.5% 
0.9861 1.8632 
```

> A product of two estimates is **not normally distributed** → no textbook standard error → **bootstrap** (Chapter 3): resample users, re-estimate, look at the distribution. Report **total, direct, indirect** with uncertainty, plus the **share mediated**. (`mediation` package: `mediate()` does this incl. logistic outcomes.)

The catch: nobody randomized the mediator

Live demo: book §6.7 — mediation widget, switch on the **unobserved confounder of M and Y**

https://wu-rds.github.io/MA2026/06-mechanisms.html#sec-mech-mediation

Screenshot placeholder: the mediation widget
`img/ch06/widget_mediation.png`

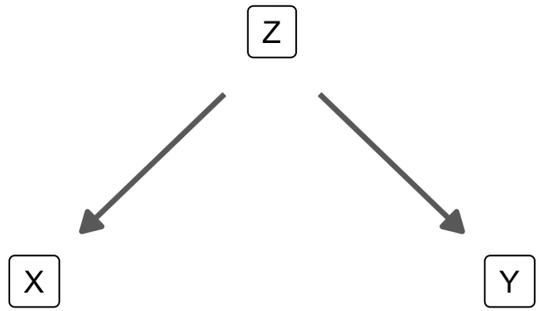
> Randomization identifies the **total** effect. The **split** needs one more assumption: **no unobserved confounder of mediator and outcome** (curious users discover more *and* listen more). Untestable — and the same problem as Chapter 5, one step down the chain.

What helps: measure likely confounders of M; **manipulate the mediator** (Smart Mix with vs. without discovery).

6.8 Good and bad controls

Fork, pipe, collider

Fork (confounder)



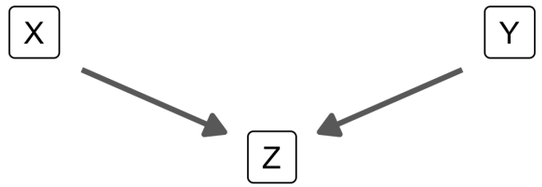
adjust for Z

Pipe (mediator)



do NOT adjust for Z
(unless you want the direct effect)

Collider (common effect)



do NOT adjust for Z

Fork = confounder → adjust (Ch. 5). **Pipe** = mediator → adjusting gives the *direct* effect, not the total. **Collider** = common effect → adjusting (or selecting on it) **creates** a spurious association. Draw the diagram **before** the analysis.

6.9 Your group project

Moderation and mediation with 120 respondents

Add a moderator?

- Only if central to your topic and you can state the **expected direction** in advance
- Measure it **before** the manipulation
- `outcome ~ condition * moderator`
- 60 per condition → only **large** interactions detectable
- A 2 × 2 spreads you over cells of 30

Measure a mediator?

- One or two plausible mechanisms, measured **after** the manipulation (perceived quality, trust, attention)
- Cheap — and tells you **where the chain broke** if the manipulation did not work
- Report total / direct / indirect **with the caveats**

> State what was **planned** and what was **exploratory**. “No evidence that the effect differs” ≠ “the effect is the same”.

Learning check

1. `smart_mix` = 5.0, `smart_mix:premium` = −3.1. Effect for free users? For Premium? What does the interaction's *p*-value test?
2. 2 × 2: intercept 40, `framingYearly` −5, `trialYes` +8, `framingYearly:trialYes` +6. The four cell means? Where does the trial help more?
3. A loyalty program's total effect on spending, with "store visits" as a control — fork, pipe, or collider? Consequence?
4. 25 segments tested, "works particularly well for returning customers on tablets" (*p* = .02). Problem — and what next?

Next session

Module 8 · Survey design — *from research question to questionnaire*

- Measuring constructs: items, scales, reliability
- Embedding your experiment in a Qualtrics survey
- Questionnaire structure, pretesting, data collection

> Bring your group's **research question and planned manipulation** — we will work on your questionnaires in class.